Boules A. Marzouk

Software Engineer • Platform, backend and Linux infrastructure
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Experience

Procore Technologies Feb 2026 – Present
Software Engineer, Internal Developer Platform *Remote*

- Developed and documented E2ETR, an end-to-end test runner on Argo Workflows that runs tests for 60+ services without blocking their pipelines.
- Halved a service's CI time (10 to 5 minutes) by moving repeated steps into a shared, cached builder.
- Redesigned CI failure alerting so teams are paged only for blocking failures.
- Serve on the global on-call rotation for the main monolith's CI pipelines, reverting breaking changes or fixing the tooling when pipelines fail.
- Maintain SonarQube and build deprecation reports that keep version updates in sync across the monorepo.
- Built a Node.js plugin for Claude Code and Cursor that cut main-context token usage by 46–99.8%, with fail-open hooks and 66 automated tests.

Siemens Digital Industries Software Feb 2025 – Dec 2025
Software Engineer *Cairo, Egypt (Hybrid)*

- Refactored a critical backend API, reducing its execution time by 80%.
- Helped design the gRPC contracts connecting 30+ internal services.
- Designed, with the tech lead, the data model of the service that became the entry point of our whole solution.
- Built an internal debugging tool still used by 40+ engineers, cutting issue-resolution time by about 30%.
- Resolved production incidents for two major enterprise accounts, 90% of them within SLA.

Xpert Key Oct 2024 – Feb 2025
Associate Software Engineer *Remote (UAE-based team)*

Open Source

maat | *Go, snapcraft, AppArmor, systemd, polkit, LXD* | snapcraft.io/maat Oct 2026 – Present

- Built an MCP server in Go that lets AI agents diagnose Ubuntu machines without changing them; it ships as a strict snap for amd64 and arm64 with no network access.
- Fixes are typed plans that a person applies through polkit, recorded in a hash-chained audit trail; remote machines are reached over SSH with forced-command keys.
- Found that snapd's `network-observe` interface permits outbound connections, which its docs didn't say, and got the fix merged into canonical/snap-docs (PR #679).

ETMRN | *TypeScript, Node.js, NestJS, WebSockets, RabbitMQ* Side project

- Co-building with my mentor the backend of a marketplace connecting clients with fitness coaches, nutritionists and physical therapists: matching, role-based workflows, real-time messaging, and integrations with three payment providers (Paymob, Kashier, Fawry).

Education

Ain Shams University Cairo, Egypt
B.Sc. in Computer Science; GPA 3.6/4.0, ranked 50th of 1,000+ students *2026*

- Ranked 850th of 689,000 nationwide (top 0.12%) in the Egyptian secondary certificate, Mathematics track.

Technical Skills

Languages: Go, Ruby, TypeScript (Node.js, NestJS), Python, C, SQL, Bash
Infrastructure: Linux (Ubuntu), systemd, AppArmor, snapcraft, LXD, Docker, Kubernetes (k3s), Argo Workflows
CI/CD and cloud: AWS, CircleCI, Artifactory, SonarQube, Git, gRPC, on-call and incident response
Data and messaging: PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, AWS SQS/SNS